
Gauge-Invariant Loop and Reservoir Completions for Real-Time Lattice Scalar Electrodynamics

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Abstract

Real-time variational lattice gauge theory can conserve charge exactly while remaining physically incomplete: geometric fidelity does not supply quantum loops, and a prescribed field does not carry a source-energy ledger. We separate these three layers—discrete geometry, effective-field content, and system closure—in scalar electrodynamics. The same vertex–edge implementation first reproduces the cold-plasma transverse dispersion with a 0.494584% median ridge error and the transition of a one-dimensional gamma-ray pulse from a $0.132143 E_c$ wake to a $1.35235 E_c$ supercritical field. It then recovers Aharonov–Bohm holonomy and the lowest scalar Landau level with finest-grid relative frequency errors 9.468×10^{-6} and 1.826×10^{-5} , respectively. Two converged counterexamples expose content rather than discretization errors. At zero scalar occupation the tree-level theory yields $\Delta n = 0$ in a magnetized vacuum although scalar QED predicts $\Delta n = -4.227 \times 10^{-4}$; adding the one-loop scalar Euler–Heisenberg face action gives -4.227×10^{-4} . A prescribed cavity pump produces a resolution-independent 0.0535519 energy defect that is exactly the omitted external work; promoting the source to a minimally coupled oscillator reduces the extended-energy drift to 2.837×10^{-7} . Both completions retain gauge symmetry, charge conservation, and the Bianchi identities. The results establish a practical rule: variational structure protects the identities of the action actually discretized, but neither repairs missing operators nor closes an open physical system.

1 Introduction

Plasma dynamics ceases to admit a clean kinetic description when field gradients approach a Compton wavelength, pair production appreciably depletes the driver, or collective and relativistic-quantum scales overlap. Semiclassical source terms can be successful in separated regimes [Berezhiani et al., 1992, Kluger et al., 1998, Schmidt et al., 1998, Roberts et al., 2002, Hebenstreit et al., 2008, 2009], but an imposed pair source does not by itself enforce simultaneous energy–momentum exchange among matter and fields. This limitation matters in laser-generated pair plasmas [Liang et al., 1998, Gahn et al., 2000, Liang et al., 2015, Sarri et al., 2015], where cascades, wakefields, and backreaction coexist [Duclos et al., 2011, Nerush et al., 2011, Ridgers et al., 2012, Di Piazza et al., 2012, Fedotov et al., 2023].

Real-time lattice gauge theory offers a complementary description. In the high-occupation, weak-coupling regime, the Schwinger–Keldysh theory [Schwinger, 1961, Keldysh, 1965] admits a classical-statistical truncation [Aarts and Berges, 2002, Mueller and Son, 2004, Jeon, 2005, Berges and Gasenzer, 2007, Berges et al., 2014, Gelis, 2015]. An ensemble of classical fields then captures tree-level dynamics without a particle deposition step. Shi *et al.* discretized scalar QED on a Yee-type spacetime complex, placing charged fields on vertices and gauge one-forms on edges [Shi et al., 2018]. Wilson lines make the discrete action locally U(1)-invariant; charge continuity follows as a discrete Noether identity, while $F = dA$ makes $dF = 0$ algebraic. This construction is closely related to gauge-compatible finite-element and variational methods [Christiansen and Halvorsen,

2011, Christiansen, 2023, Marsden and West, 2001, Hairer et al., 2006] and to structure-preserving plasma algorithms [Squire et al., 2012, Xiao et al., 2013, 2015, Qin et al., 2016].

Exact identities can, however, invite an unsafe inference: a scheme may be geometrically exact for an action that omits the relevant physics. The classical-statistical action has no vacuum loop at zero occupation [Aarts and Smit, 1999, Gelis and Tanji, 2013]; the Maxwell–Klein–Gordon energy theorem also presumes that every energy-bearing source is varied. Neither defect is removed by spatial or temporal refinement. Conversely, before calling a discrepancy fundamental one must establish that the implementation reproduces known coupled dynamics and that the discrepancy survives convergence.

Here we make that separation operational. We verify the original cold-plasma and laser-slab demonstrations in one implementation, then probe flat connections on a ring and quantized magnetic flux on a torus. Those regimes exercise gauge covariance and non-trivial matter–link coupling without repeating an unmagnetized dispersion test. We next isolate two obstructions. Magnetized vacuum propagation is blind to scalar loops when $\phi = 0$, and a continuously prescribed pump has no dynamical energy reservoir. We derive and implement an action-level completion for each: the weak-field scalar Euler–Heisenberg operator [Heisenberg and Euler, 1936, Weisskopf, 1936, Dunne, 2005] and a minimally coupled source coordinate. The central finding is not that variational lattice QED fails, but that its guarantee has a precise domain: it preserves the symmetries of the chosen closed action and nothing beyond them.

2 Methodology

2.1 Vertex–edge scalar electrodynamics

In natural units, the continuum action for charged scalar species is

$$S[\phi, A] = \int d^4x \left[(D_\mu \phi)^* D^\mu \phi - m^2 |\phi|^2 - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} \right], \quad D_\mu = \partial_\mu - iqA_\mu. \quad (1)$$

For multiple species the matter terms and currents are summed. On a rectangular spacetime complex, ϕ_v is vertex-centered and A_e is an oriented edge one-form. With

$$U_e = \exp(-iq\Delta_e A_e), \quad (D^<\phi)_e = \frac{U_e \phi_{e+} - \phi_{e-}}{\Delta_e}, \quad (2)$$

and face curvature $F_f = (dA)_f$, our discrete action is

$$S_d = \sum_c \Delta V_c \left[\eta^{\mu\nu} (D_\mu^<\phi)_e^* (D_\nu^<\phi)_e - m^2 |\phi_v|^2 + \frac{1}{2} (E_f^2 - B_f^2) \right]. \quad (3)$$

This is the non-compact face-field gauge action of Ref. [Shi et al., 2018], not a plaquette-cosine surrogate [Wilson, 1974]. Under $\phi_v \mapsto e^{iq\alpha_v} \phi_v$ and $A_e \mapsto A_e + (\alpha_{e+} - \alpha_{e-})/\Delta_e$, every term in Eq. (3) is invariant.

We use temporal gauge $A_0 = 0$. Variation gives the explicit leapfrog

$$\phi^{n+1} = 2\phi^n - \phi^{n-1} + \Delta t^2 (\Delta_A \phi^n - m^2 \phi^n), \quad (4)$$

$$\mathbf{A}^{n+1} = 2\mathbf{A}^n - \mathbf{A}^{n-1} + \Delta t^2 [-\nabla_h \times \nabla_h \times \mathbf{A}^n + \mathbf{J}^n], \quad (5)$$

where every term of Δ_A is parallel transported. On an x -link,

$$J_{i+1/2}^n = -\frac{2q}{\Delta x} \text{Im} \left[\bar{\phi}_{i+1}^n e^{iq\Delta x A_{i+1/2}^n} \phi_i^n \right], \quad \rho_i^{n+1/2} = -\frac{2q}{\Delta t} \text{Im}(\bar{\phi}_i^{n+1} \phi_i^n). \quad (6)$$

Equations (4)–(6) come from the same action; mixing them with a continuum or cell-centered current would forfeit the argument. Their discrete divergence gives

$$\frac{\rho^{n+1/2} - \rho^{n-1/2}}{\Delta t} = \nabla_h \cdot \mathbf{J}^n, \quad (7)$$

so periodic total charge is constant to roundoff. Because $F_f = dA_e$, the magnetic divergence and Faraday identity follow from $d^2 = 0$, rather than from a separately advanced equation. We monitor a positive midpoint shadow energy. At finite coupling its deviation from the continuum Noether energy is $\mathcal{O}(\Delta t^2 + \Delta x^2)$, while charge and the Bianchi identities are exact identities of the update.

Table 1: Pre-result diagnostic protocol. The failure clauses for the last two rows are definitions fixed before inspecting their production outputs.

Physical setting	Primary diagnostic	Sharp prediction	Fixed criterion
Flux-threaded ring	Gauge invariance	$\omega^2 = m^2 + (2\pi\ell/L - q\bar{A})^2$; gauge-related histories coincide	finest frequency error $\leq 0.5\%$; gauge-orbit mismatch $\leq 5 \times 10^{-12}$
Scalar Landau level	Non-trivial ϕ - A coupling	$\omega_0^2 = m^2 + qB $	finest frequency error $\leq 1\%$; plaquette closure $\leq 5 \times 10^{-13}$
Magnetized vacuum	Dispersion fidelity	one-loop $\Delta n = n_{\parallel} - n_{\perp}$ from Eq. (10)	fundamental failure if extrapolated tree-level $ \Delta n < 10^{-5}$ while predicted $ \Delta n > 3 \times 10^{-4}$; fixed error $\leq 3 \times 10^{-5}$, energy drift $\leq 2 \times 10^{-5}$
Driven cavity	Energy conservation	$\Delta\mathcal{U} = W_{\text{ext}}$ for a prescribed pump; conserved extended energy after source completion	fundamental failure if nominal energy changes $> 10^{-2}$ at every resolution while the work ledger closes within 5×10^{-4} ; fixed energy drift $\leq 2 \times 10^{-4}$, charge drift $\leq 5 \times 10^{-11}$

2.2 Prospective predictions and diagnostic separation

Table 1 states the predictions, primary diagnostic, and acceptance or failure threshold fixed in `code/protocol.json` before production. The tolerances have different origins. Frequency tolerances exceed the measured second-order truncation envelope; exact-symmetry tolerances are roundoff-scaled; the vacuum threshold separates a finite one-loop effect from zero; and the pump threshold is more than two orders above the converged integrator error. Each setting uses at least three resolutions. No two settings assign the same primary bug class.

The ring has a constant flat connection $\bar{A}$ but nonzero holonomy. A periodic mode $e^{2\pi i \ell x/L}$ therefore has the continuum frequency shown in Table 1; its lattice reference is obtained by $p^2 \mapsto 4 \sin^2(p\Delta x/2)/\Delta x^2$. A periodic gauge function $\alpha(x)$ changes every local link but not the holonomy. This is primarily a gauge test because the entire field history, not merely a scalar spectrum, must transform covariantly.

For both scope settings we retain the complex coefficient $c_h(t) = \langle u_h, \phi_h(t) \rangle_h$. The sharp continuum frequencies therefore also predict the parameter-free normalized trajectories

$$\frac{c_h(t)}{c_h(0)} = \exp(-i\omega t), \quad (8)$$

which we compare at every stored time level. The corresponding discrete frequency is used only as a cross-check that separates accumulated $\mathcal{O}(\Delta x^2 + \Delta t^2)$ phase error from amplitude or mode-shape error; it does not replace the pre-stated continuum criterion.

For a square torus carrying N_Φ flux quanta,

$$qBL_xL_y = 2\pi N_\Phi. \quad (9)$$

The scalar spectrum is $\omega_n^2 = m^2 + (2n+1)|qB|$ [Al-Hashimi and Wiese, 2009, Zak, 1964]. The lowest frequency is a coupling diagnostic: it is generated solely by Wilson phases around nontrivial plaquettes, while a free finite difference cannot produce it.

2.3 Loop completion of the empty magnetized vacuum

If $\phi = 0$, Eqs. (4)–(5) reduce exactly to source-free Maxwell theory. No ensemble size can create a vacuum polarization loop from that deterministic state. In weak, slowly varying fields, integrating out N_s charged scalars gives

$$S_{\text{EH}} = S_d + \sum_c \Delta V_c [a(\mathbf{E}^2 - \mathbf{B}^2)^2 + b(\mathbf{E} \cdot \mathbf{B})^2], \quad a = \frac{7N_s\alpha^2}{360m^4}, \quad b = \frac{4N_s\alpha^2}{360m^4}, \quad (10)$$

with $\alpha = q^2/(4\pi)$ [Weisskopf, 1936, Schwinger, 1951, Dunne, 2005]. We place the correction on faces, so it depends only on F_f . Its constitutive equations are

$$\mathbf{D} = \mathbf{E} + 4a(\mathbf{E}^2 - \mathbf{B}^2)\mathbf{E} + 2b(\mathbf{E} \cdot \mathbf{B})\mathbf{B}, \quad (11)$$

$$\mathbf{H} = \mathbf{B} + 4a(\mathbf{E}^2 - \mathbf{B}^2)\mathbf{B} - 2b(\mathbf{E} \cdot \mathbf{B})\mathbf{E}, \quad (12)$$

$$\nabla_h \cdot \mathbf{D} = \rho, \quad \Delta_t \mathbf{D} = \nabla_h \times \mathbf{H} - \mathbf{j}. \quad (13)$$

For propagation perpendicular to $\mathbf{B}_0 = B_0 \hat{z}$, linearization gives

$$\epsilon_{\perp} = 1 - 4aB_0^2, \quad \mu_{\parallel}^{-1} = 1 - 12aB_0^2, \quad (14)$$

$$\epsilon_{\parallel} = 1 - 4aB_0^2 + 2bB_0^2, \quad \mu_{\perp}^{-1} = 1 - 4aB_0^2, \quad (15)$$

and $n_{\perp, \parallel} = \sqrt{\epsilon_{\perp, \parallel} / \mu_{\perp, \parallel}^{-1}}$. We implement the quadratic probe action obtained by this linearization; it is the Hessian of Eq. (10), not an imposed phase velocity. Gauge invariance, Gauss charge, $dF = 0$, and autonomous effective-action energy are preserved. The strict tree-level truncation is intentionally relaxed, and the local fix is not claimed for ω or gradients comparable to m [Adler, 1971, Karbstein, 2017, Hattori and Itakura, 2013, Gies and Karbstein, 2017].

2.4 Dynamical completion of a prescribed source

A homogeneous transverse link reduces the matter frequency to

$$\Omega_{\phi}^2(A) = m^2 + \frac{4}{h^2} \sin^2\left(\frac{qhA}{2}\right). \quad (16)$$

Prescribing $A(t)$ makes the reduced action explicitly time-dependent. Thus a changing matter-plus-field energy is not an integrator failure: $d\mathcal{U}/dt = J_{\text{ext}}\dot{A}$. To close the system, we take a charged source coordinate X whose world-line minimal coupling reduces in temporal gauge to

$$L_{\text{closed}} = \frac{1}{2}|\dot{\phi}|^2 - \frac{1}{2}\Omega_{\phi}^2(A)|\phi|^2 + \frac{1}{2}\dot{A}^2 + \frac{M}{2}(\dot{X}^2 - \Omega_s^2 X^2) + gA\dot{X}. \quad (17)$$

The last term changes by a boundary term under a gauge transformation of the parent world-line action. Variation yields

$$\ddot{\phi} + \Omega_{\phi}^2(A)\phi = 0, \quad (18)$$

$$\ddot{A} = -\frac{1}{2}\partial_A \Omega_{\phi}^2(A)|\phi|^2 + g\dot{X}, \quad (19)$$

$$M\ddot{X} + M\Omega_s^2 X = -g\dot{A}. \quad (20)$$

The conserved extended energy is

$$\mathcal{U}_{\text{ext}} = \frac{1}{2}\left(|\dot{\phi}|^2 + \Omega_{\phi}^2(A)|\phi|^2 + \dot{A}^2 + M\dot{X}^2 + M\Omega_s^2 X^2\right). \quad (21)$$

We advance the Hamiltonian form with the implicit midpoint variational integrator. The fix preserves scalar U(1) charge, gauge invariance, and extended energy to second-order integration error. It necessarily relaxes the exactly prescribed pump waveform and conservation of the open matter-gauge subsystem.

3 Experimental Setup

3.1 Common implementation and verification runs

All production runs use double precision for evolution and five ensemble seeds $\{1729, 1733, 1741, 1753, 1777\}$, except the linear-spectrum verification, which uses eight seeds. Field histories for one representative seed and diagnostic histories for all seeds are retained. Cold initial data use the discrete rest frequency $2 \arcsin(m\Delta t/2)/\Delta t$, with amplitude chosen so Eq. (6) cancels the immobile ion density exactly at initialization.

For the unmagnetized verification we set $m = 1$, $q = 0.3$, $\omega_p = 0.85m$, $N_x = 256$, $m\Delta x = 0.08$, $m\Delta t = 0.02$, and 8192 steps. White gauge noise of standard deviation $10^{-4}m$ excites both longitudinal and transverse branches. The transverse ridge is compared to $\omega^2 = \omega_p^2 + k^2$; the fixed tolerance is a 5% median ridge error, dominated by the finite record bin width, and 5×10^{-12} charge drift.

The laser verification retains the original dimensionless density $n_0 = m^3$, $q = 0.3$, carrier $\omega_0 = 0.7m$, and intensities $a = 1$ and 16 [Shi et al., 2018]. A reduced but fully resolved periodic box has $N_x = 1600$, $Lm = 160$, $m\Delta x = 0.1$, $m\Delta t = 0.02$, a width-50/ m plasma plateau, Gaussian

ramp $10/m$, and pulse duration $10/m$. Five statistically equivalent smooth phase perturbations are advanced for 6000 steps. This is a physics reproduction, not a digitization of the published images: it tests the same subcritical wake/supercritical transition in a smaller box. The pre-stated criterion requires $\max |E_x| < E_c = m^2/q$ for $a = 1$, $> E_c$ for $a = 16$, and a high-field positive-charge yield at least five times larger.

3.2 Holonomy, magnetic flux, and boundary configurations

The ring uses $m = 1$, $q = 0.3$, $L = 20$, mode $\ell = 2$, and $\bar{A} = 0.7$, at $N_x = \{64, 96, 128, 192\}$ with $\Delta t = 0.2\Delta x$. We compare a constant-link history with one generated by a two-harmonic periodic $\alpha(x)$. The magnetic torus has $L_x = L_y = 12$, $N_\Phi = 4$, and $N = \{24, 32, 48, 64\}$ per side. A sparse Hermitian solve initializes the lowest covariant-Laplacian eigenvector; the subsequent frequency is measured from its complex modal phase. The trajectory overlays use every stored time level of the finest-grid, seed-1729 history, while the convergence points and acceptance decisions use all five seeds.

The vacuum run uses four scalar species, $q = m = 1$, $B_0 = 0.5m^2$, a length-20 box, and the fundamental probe $k = 2\pi/20$, safely below the scalar mass for the local derivative expansion. We evolve $N_x = \{80, 120, 160, 240\}$ with $\Delta t = 0.2\Delta x$, both without and with Eq. (10). Random global probe phases form the five-member ensemble; their vanishing statistical spread is expected for a linear deterministic mode.

The driven cavity takes $m = 1$, $q = 0.3$, $h = 0.5$, and a prescribed $A(t) = 2 \sin t$ for 12 periods. Resolutions are $\Delta t = \{0.02, 0.01, 0.005\}$. The closed model uses $M = 1$, $g = 0.6$, and chooses $\Omega_s = \sqrt{1 - g^2/M}$, so the uncoupled source-gauge normal mode has unit frequency. For the prescribed case we use the exact discrete work identity associated with Eq. (4), avoiding a continuum/discrete centering mismatch. All configurations, seeds, timestamps, and complete diagnostic arrays are enumerated in the log manifest.

4 Experimental Results

4.1 Verification foundation

Figure 1 establishes that the code used to diagnose fundamental obstructions first reproduces the published coupled regimes. The transverse spectrum tracks the analytic branch over the resolved band with a 0.494584% median ridge error. Total scalar charge changes by at most 1.340×10^{-14} , the total-energy shadow diagnostic by 1.265×10^{-7} , and the maximum Gauss residual is recorded in the accompanying run log.

The laser comparison crosses the pre-stated physical threshold. The low-intensity ensemble reaches $\max |E_x|/E_c = 0.132143 \pm 5.483 \times 10^{-5}$, while the high-intensity ensemble reaches $1.35235 \pm 1.892 \times 10^{-4}$. The latter develops positive-charge domains only after the pulse and reflected/plasma waves overlap, and its peak yield is 1.038×10^3 times the low-field value. Field energy is transferred to scalar energy with total-energy drift no larger than 2.787×10^{-4} ; charge drift remains 5.838×10^{-15} . The reproduction therefore rules out a globally incorrect current sign, broken Gauss update, or absent nonlinear transverse coupling as explanations for the boundary results below.

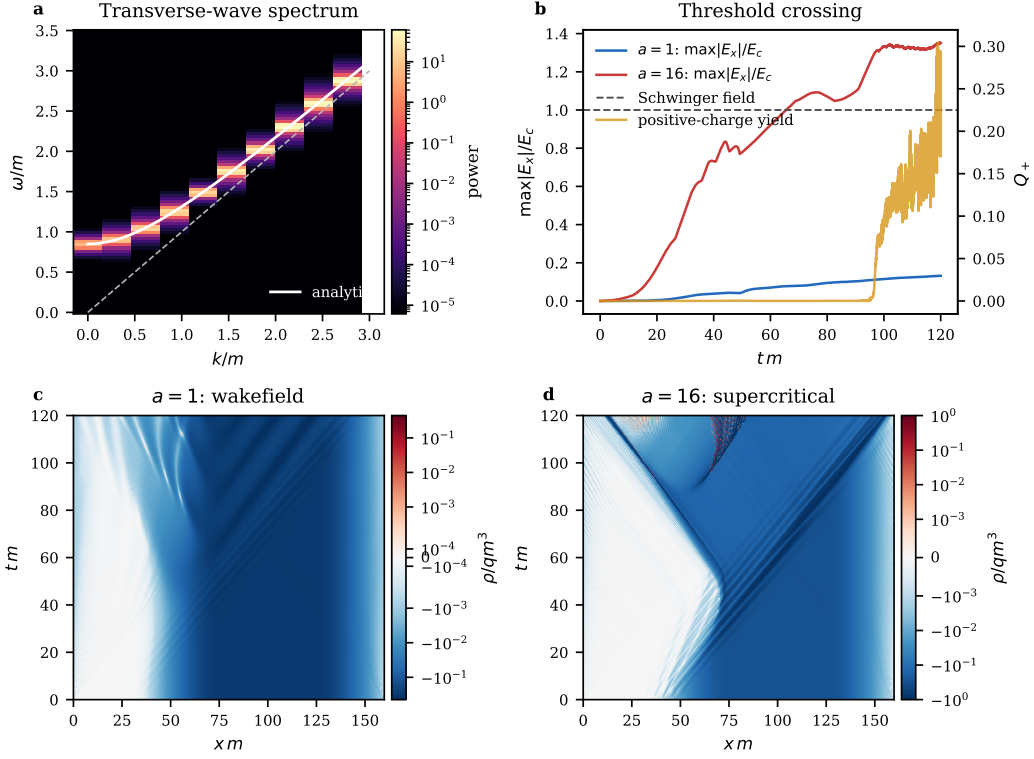


Figure 1: **Verification in the implementation used throughout.** (a) Ensemble transverse-electric spectrum and $\omega^2 = \omega_p^2 + k^2$ overlay. (b) Longitudinal wake amplitude and positive-charge yield. (c,d) Representative scalar charge histories for $a = 1$ and $a = 16$; the high-field case crosses E_c and develops opposite-sign domains. Source arrays: `paper_sec_III1.spectra.npz` and `paper_sec_III2.a1` and `paper_sec_III2.a16` field logs.

Table 2: Primary results. Every entry is generated from the JSON/NPZ run logs; $\|\Delta Q/Q\|$ and $\|\Delta \mathcal{U}/\mathcal{U}\|$ are maximum observed relative deviations. For the driven boundary row, the energy column is the converged defect of the unclosed model; the fixed extended-energy value appears in the outcome.

Test	Category	Regime probed	Prediction	Tolerance / failure criterion	Outcome	$\ \Delta Q/Q\ $	$\ \Delta \mathcal{U}/\mathcal{U}\ $
Cold-plasma spectrum (Shi et al. Sec. III.1)	verification	unmagnetized 1D dispersion	$\omega^2 = \omega_p^2 + k^2$	median ridge error $\leq 5\%$	0.494584% median error; passed	1.340×10^{-14}	1.265×10^{-7}
Gamma-ray slab (Shi et al. Sec. III.2)	verification	1D laser plasma, $a = 1, 16$	wakefield below and charge production above E_c	$a = 1 < E_c, a = 16 > E_c$, yield ratio > 5	0.132143 vs. 1.35235 E_c ; yield ratio 1.038×10^3	5.838×10^{-15}	2.787×10^{-4}
Flux-threaded ring	scope	flat connection with nonzero holonomy	$\omega^2 = m^2 + (2\pi\ell/L - q\bar{A})^2$	frequency error $\leq 0.5\%$, gauge mismatch $\leq 5 \times 10^{-12}$	fine error 9.468×10^{-6} ; gauge mismatch 3.929×10^{-13}	1.263×10^{-14}	2.286×10^{-10}
Scalar Landau level	scope	quantized 2D magnetic flux	$\omega_0^2 = m^2 + qB $	frequency error $\leq 1\%$, plaquette error $\leq 5 \times 10^{-13}$	fine error 1.826×10^{-5} ; plaquette 3.940×10^{-15}	1.197×10^{-15}	4.431×10^{-14}
Magnetized vacuum	boundary	zero-occupation one-loop response	$\Delta n = -4.227 \times 10^{-4}$	failure: tree $ \Delta n < 10^{-5}$ while prediction $> 3 \times 10^{-4}$; fixed error $\leq 3 \times 10^{-5}$	tree 0; EH fix -4.227×10^{-4} (error 3.834×10^{-12})	0	3.952×10^{-14}
Driven cavity	boundary	sustained external strong-field pump	$\Delta \mathcal{U} = W_{\text{ext}}$; closed $\mathcal{U}_{\text{ext}}$ constant	failure: defect $> 10^{-2}$, ledger $< 5 \times 10^{-4}$; fixed drift $\leq 2 \times 10^{-4}$	defect 0.0535519; ledger 2.689×10^{-13} ; reservoir fix 2.837×10^{-7}	8.704×10^{-13}	0.0535519

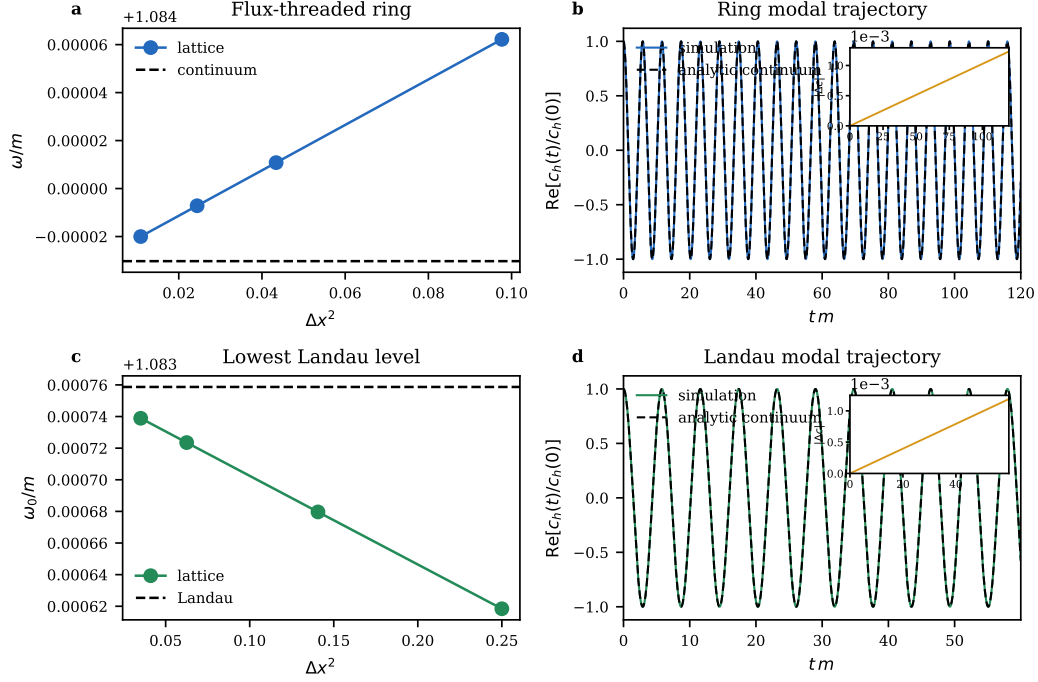


Figure 2: **Gauge and magnetic regimes beyond the original demonstrations.** (a) Ring frequency versus Δx^2 , with the continuum holonomy value. (b) Finest-grid ring modal trajectory (solid) against Eq. (8) (dashed). (c) Lowest scalar Landau frequency versus Δx^2 . (d) Finest-grid Landau trajectory against the analytic lowest-level prediction. Insets show the full-record complex discrepancy; error bars from the five phase/noise seeds are smaller than the markers.

4.2 Flat connections and quantized magnetic flux

The ring result in Fig. 2 converges linearly in Δx^2 to the holonomy prediction $\omega = 1.08397$. The finest relative error is 9.468×10^{-6} . More stringently, locally different link representatives generate histories related by the vertex phase to within 3.929×10^{-13} . This comparison would expose a link orientation, current centering, or boundary-wrap error even if the fitted frequency happened to be correct. Charge drift is 1.263×10^{-14} and energy drift 2.286×10^{-10} . Over the full stored trajectory, Fig. 2b gives $\max_t |c_h(t)/c_h(0) - e^{-i\omega t}| = 1.232 \times 10^{-3}$, with RMS 7.110×10^{-4} . Replacing ω by the independently registered finite-grid frequency reduces the maximum to 4.188×10^{-13} , identifying the continuum residual as the expected accumulated phase error.

On the torus, every plaquette carries the same phase and closes within 3.940×10^{-15} . The measured lowest mode converges to $\omega_0 = 1.08376$, with finest relative error 1.826×10^{-5} . Its full-record trajectory comparison in Fig. 2d has maximum complex discrepancy 1.187×10^{-3} and RMS 6.857×10^{-4} ; the discrete-frequency cross-check reduces the maximum to 2.415×10^{-14} . The localized density is gauge-dependent, whereas the eigenvalue and its N_Φ -fold low-level structure are not. Because the shift disappears when the Wilson phases are removed, this test directly exercises non-trivial ϕ - A coupling rather than another unmagnetized free-field dispersion.

4.3 A converged absence of vacuum polarization

The tree-level vacuum produces no birefringence at any resolution, hence $\Delta n \rightarrow 0$, while Eq. (10) predicts -4.227×10^{-4} . This satisfies the failure signature in Table 1. The discrepancy cannot be a resolution artifact: the two polarizations obey identical Maxwell updates for every Δx , and their extrapolated difference remains zero. It also cannot be assigned to a general implementation failure after the plasma-wave, laser, holonomy, and magnetic-level checks.

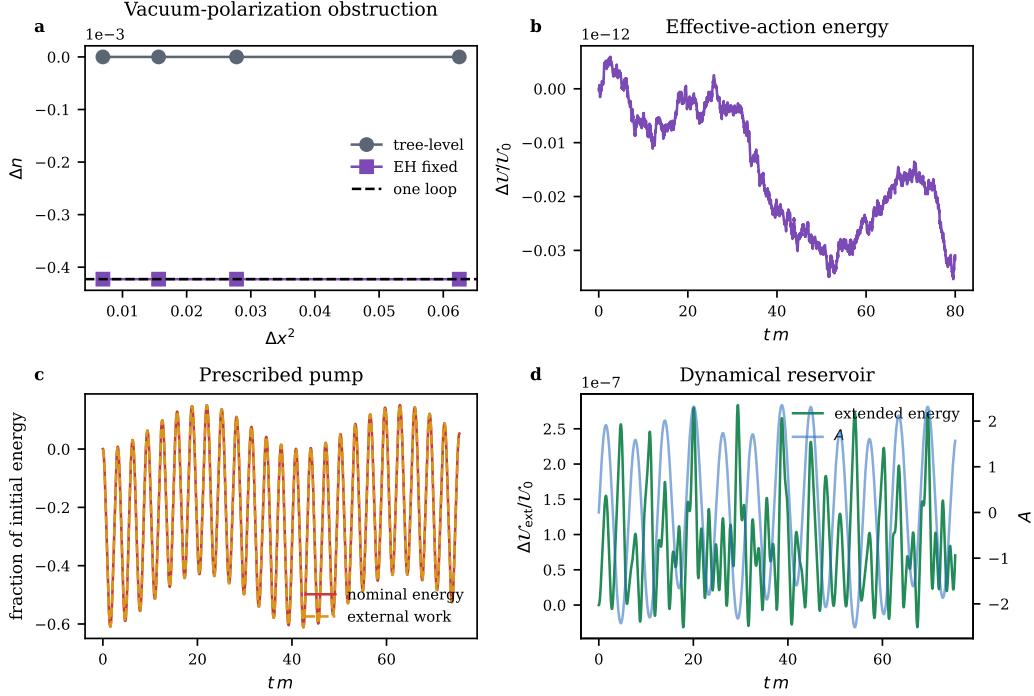


Figure 3: **Fundamental obstructions and action-level completions.** (a) Tree-level and Euler–Heisenberg birefringence versus Δx^2 ; the dashed line is the one-loop prediction. (b) Effective-action energy trace. (c) Prescribed-pump energy change coincides with omitted external work. (d) The dynamical reservoir allows A to deplete while the extended energy remains bounded.

The face-action completion resolves the discrepancy. Its extrapolated value is -4.227×10^{-4} , an absolute error 3.834×10^{-12} . The Bianchi residual is at most 1.499×10^{-15} and effective-action energy drift 3.952×10^{-14} . Scalar charge is identically zero because the test has no matter excitation; gauge invariance ensures that adding matter would retain the same Noether identity. The result is deliberately narrow: it verifies the leading local one-loop operator, not the nonlocal polarization tensor near pair threshold [Ritus, 1975, Dittrich and Reuter, 1985, Baier et al., 1975].

4.4 Closing the energy ledger of a sustained pump

Refinement does not remove the prescribed-pump defect (Fig. 3). At the finest step the nominal matter-plus-field energy changes by 0.0535519 , while the discrete external-work ledger closes to 2.689×10^{-13} . The coexistence of a finite energy defect with charge drift only 8.704×10^{-13} is diagnostic: local $U(1)$ symmetry remains intact, but time-translation symmetry is absent because the pump is not varied. This is a modeling obstruction rather than numerical heating.

With the source coordinate varied, energy flows among matter, gauge mode, and reservoir; matter gains 0.739495 of its initial energy while $\mathcal{U}_{\text{ext}}$ drifts by only 2.837×10^{-7} . The drift decreases quadratically with Δt . The exact prescribed waveform is no longer an invariant—pump depletion is the physical content of the fix. Thus the completion preserves global charge and extended energy while explicitly relaxing field-only energy conservation and externally enforced amplitude.

4.5 Convergence and conservation audit

Figure 4 collects all four new resolution studies. The two predictive spectra approach their continuum values with the expected second-order trend. The vacuum baseline stays exactly polarization-degenerate while the fixed sequence approaches the one-loop line. The prescribed-drive defect remains finite as $\Delta t \rightarrow 0$, whereas the fixed energy error falls with the midpoint integrator. Figure 5 shows a representative energy history for every physical setting. Full charge, energy-component,

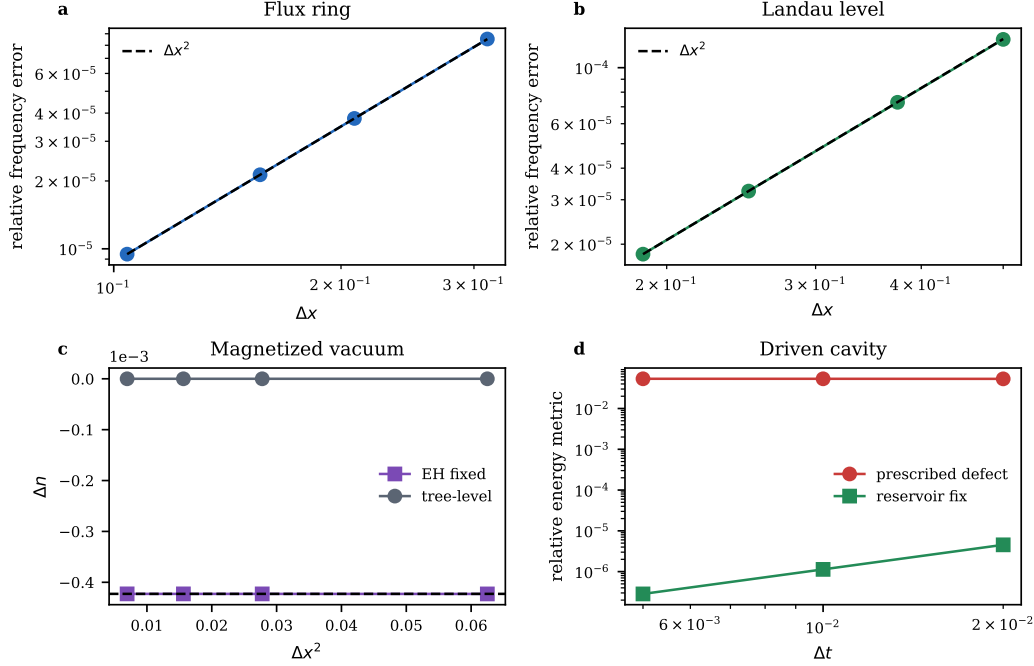


Figure 4: **Resolution separation of truncation error from missing physics.** (a,b) Frequency errors for the ring and Landau level. (c) Vacuum birefringence: the tree-level absence persists, while the effective action converges to the prediction. (d) The unclosed energy defect persists while the reservoir-fix error decreases.

Gauss, Bianchi, and observable histories—not only plotted subsamples—are retained in the NPZ logs.

5 Analysis

5.1 Three logically independent notions of fidelity

The experiments expose a hierarchy that is easy to obscure when all checks are called “validation.” First, *geometric fidelity* asks whether discrete incidence and parallel transport realize the symmetries of a chosen action. The ring, plaquette, Gauss, and charge results answer this layer. Second, *effective-theory fidelity* asks whether that action contains the operators relevant to the state. An empty classical-statistical vacuum has no charged fluctuation to circulate around a loop; exact gauge symmetry therefore coexists with exactly wrong birefringence. Third, *system-boundary fidelity* asks whether every energy-bearing component has been included in the variational principle. The driven case has the right local charge theorem but the wrong closed-system energy claim.

This hierarchy explains why refinement is decisive but not sufficient. Second-order convergence removes the discretization layer. It cannot alter the loop order of an action or turn a prescribed function into a dynamical degree of freedom. Conversely, simply observing a nonconvergent or inaccurate quantity would not establish a fundamental limitation; the successful reproductions and the orthogonal gauge/coupling diagnostics are what make the two negative results interpretable.

5.2 Why the two completions preserve structure

Both repairs act before variation. The Euler–Heisenberg operator is built only from face curvatures. It changes the constitutive map $(E, B) \mapsto (D, H)$, but not the cochain identity $d^2 = 0$ or the scalar phase symmetry. The reservoir completion adds a world-line degree of freedom through minimal coupling; the velocity-linear term cancels from Eq. (21), leaving a transparent Noether ledger. In

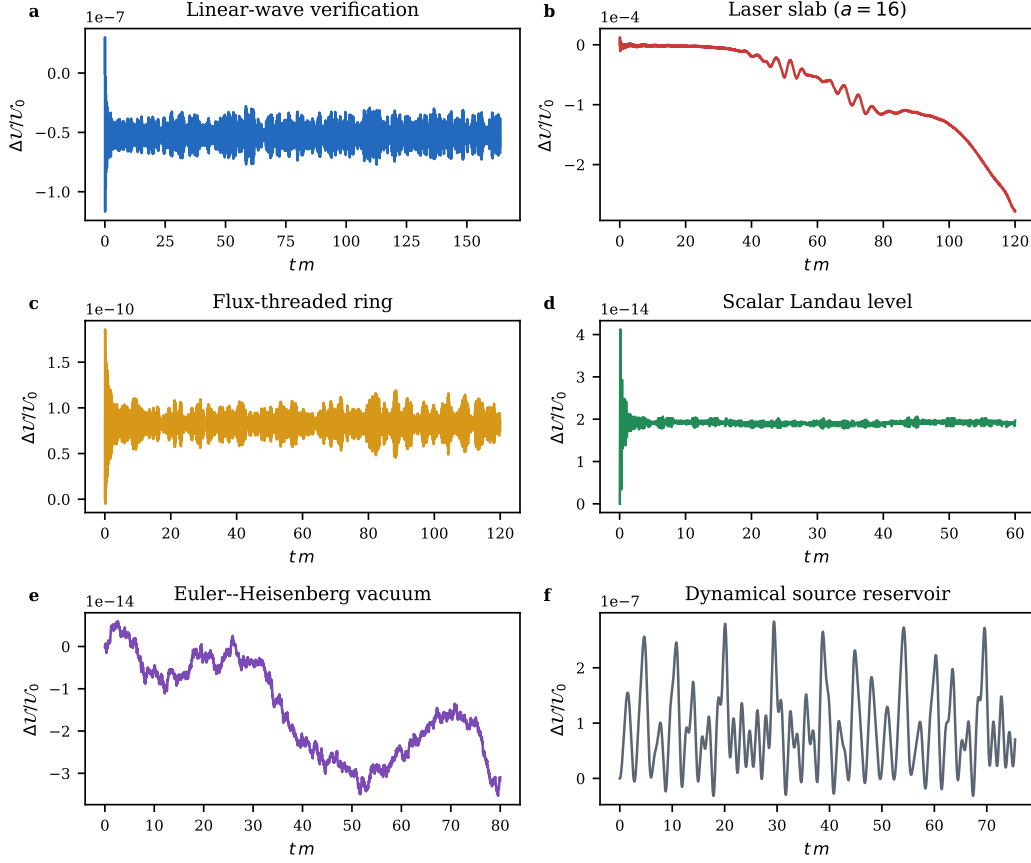


Figure 5: **Conservation traces for one realization of all six settings.** Panels show the relative midpoint/shadow-energy deviation for (a) linear waves, (b) the $a = 16$ laser slab, (c) the flux ring, (d) the Landau mode, (e) the loop-completed vacuum, and (f) the dynamical-source completion. Different ordinate multipliers are shown on each panel. Corresponding charge histories underlie Table 2.

neither case is a correction inserted directly into an update equation. That action-first property is the transferable design principle.

The fixes also differ in what they concede. Loop completion abandons the tree-level approximation but preserves a closed gauge-field action. Reservoir completion abandons a prescribed waveform but preserves the tree-level matter model. Treating either concession as a numerical nuisance would reintroduce the original inconsistency: enforcing the Maxwell light cone would erase vacuum polarization, and enforcing the pump amplitude would erase depletion.

5.3 Implications and limitations

The work remains a scalar model. Spin, Fermi statistics, Pauli blocking, and the lattice-fermion doubling problem are absent [Nielsen and Ninomiya, 1981, Hebenstreit et al., 2013a, Kasper et al., 2014]; consequently the laser result is a verification of the scalar-QED implementation, not a quantitative prediction for an electron-positron experiment. The Euler-Heisenberg run uses four identical charged scalar species and $\alpha < 0.1$ to make a small one-loop splitting numerically resolvable. Its fundamental-mode probe is below m , but the local weak-field action does not cover pair-producing electric backgrounds, rapid gradients, dichroism, or higher loops [Dunne and Schubert, 2005, Gies and Karbstein, 2017].

The ring and Landau tests use static background links, so they test matter propagation and gauge structure rather than magnetic backreaction. The laser reproduction is one-dimensional, uses a

shorter pulse and smaller ensemble than Ref. [Shi et al., 2018], and cannot represent transverse separation of pairs. Our positive midpoint energy is a convergent shadow diagnostic, not an exact finite-step interacting energy; this is why we report its measured drift rather than asserting machine conservation. The dynamical source is a minimal single-mode completion, not a model of a laboratory laser chain. Finally, a local Euler–Heisenberg term avoids storing a causal nonlocal polarization kernel. Extending the same action-first construction to a retarded auxiliary-field representation, and to Wilson or staggered spinors, is the natural next step.

6 Conclusion

Variational lattice scalar electrodynamics passes demanding gauge, magnetization, plasma-wave, and nonlinear laser checks, yet two failures survive refinement for reasons that the discrete geometry cannot cure. Zero-occupation classical-statistical dynamics omits vacuum loops, and a prescribed pump omits its energy reservoir. A scalar Euler–Heisenberg face action restores the former response without sacrificing gauge or Bianchi structure; a minimally coupled dynamical source restores the latter ledger without sacrificing charge. The resulting distinction—geometry, field content, and closure—provides a sharper criterion for trustworthy real-time lattice plasma simulations than conservation tests alone.

A Discrete identities and analytic references

A.1 Exact charge continuity

Varying Eq. (3) under an infinitesimal vertex phase $\delta\phi_v = iq\alpha_v\phi_v$ and the compensating edge shift gives

$$0 = \sum_v \alpha_v \left[\frac{\rho_v^{n+1/2} - \rho_v^{n-1/2}}{\Delta t} - (\nabla_h \cdot \mathbf{J}^n)_v \right].$$

Because α_v is arbitrary, Eq. (7) holds pointwise. Summation over a periodic complex cancels oriented edge currents pairwise. The reported charge errors are therefore accumulated floating-point errors, not truncation errors.

A.2 Scalar Euler–Heisenberg coefficient

The renormalized one-loop proper-time integrand for a charged scalar contains

$$\frac{(qas)(qbs)}{\sinh(qas)\sin(qbs)} = 1 + \frac{q^2 s^2}{6}(b^2 - a^2) + q^4 s^4 \left[\frac{7(a^4 + b^4)}{360} - \frac{a^2 b^2}{36} \right] + \dots$$

After the vacuum and quadratic counterterms cancel, the s^4 term and $\int_0^\infty se^{-m^2 s} ds = m^{-4}$ give

$$\mathcal{L}_{\text{EH}}^{(1)} = \frac{\alpha^2}{360m^4} [7(\mathbf{E}^2 - \mathbf{B}^2)^2 + 4(\mathbf{E} \cdot \mathbf{B})^2] \mathcal{O}(F^6, \partial^2 F^4/m^2),$$

per scalar species, which is Eq. (10).

A.3 Discrete work identity

For the prescribed reduced oscillator,

$$\phi_{n+1} - 2\phi_n + \phi_{n-1} + \Delta t^2 \Omega_n^2 \phi_n = 0.$$

Multiplication by $(\bar{\phi}_{n+1} - \bar{\phi}_{n-1})/(2\Delta t)$, followed by taking the real part, yields the exact half-step identity

$$\mathcal{U}_{n+1/2} - \mathcal{U}_{n-1/2} = \frac{1}{2}(\Omega_n^2 - \Omega_{n-1}^2) \text{Re}(\bar{\phi}_{n-1}\phi_n),$$

where

$$\mathcal{U}_{n+1/2} = \frac{1}{2} \left| \frac{\phi_{n+1} - \phi_n}{\Delta t} \right|^2 \frac{1}{2} \Omega_n^2 \text{Re}(\bar{\phi}_n \phi_{n+1}).$$

This is the external-work ledger used in the production run. A continuum midpoint quadrature was rejected in the attempts log because it mixed centerings and failed the pre-stated tolerance.

B Resolution-level results

Table 3: Scope-test convergence. Errors are relative to the continuum frequencies.

Setting	N	Δx	frequency error	charge drift	energy drift
Flux ring	64	0.3125	8.529×10^{-5}	4.612×10^{-15}	3.485×10^{-11}
	96	0.2083	3.788×10^{-5}	5.635×10^{-15}	6.206×10^{-11}
	128	0.1563	2.131×10^{-5}	1.263×10^{-14}	1.088×10^{-10}
	192	0.1042	9.468×10^{-6}	1.024×10^{-14}	2.286×10^{-10}
Landau	24	0.5000	1.294×10^{-4}	1.197×10^{-15}	4.431×10^{-14}
	32	0.3750	7.294×10^{-5}	1.196×10^{-15}	4.314×10^{-14}
	48	0.2500	3.246×10^{-5}	8.539×10^{-16}	4.161×10^{-14}
	64	0.1875	1.826×10^{-5}	1.025×10^{-15}	4.160×10^{-14}

Table 4: Boundary-test convergence. Δn_{tree} is zero at every resolution. The drive defect is the nominal open-system energy change; the fix column is extended-energy drift.

Setting	resolution	baseline failure metric	fixed metric	conservation check
Vacuum	$N = 80$	$\Delta n = 0$	-4.22860×10^{-4}	$\Delta \mathcal{U}/\mathcal{U} = 1.179 \times 10^{-14}$
	$N = 120$	$\Delta n = 0$	-4.22797×10^{-4}	$\Delta \mathcal{U}/\mathcal{U} = 1.686 \times 10^{-14}$
	$N = 160$	$\Delta n = 0$	-4.22775×10^{-4}	$\Delta \mathcal{U}/\mathcal{U} = 2.442 \times 10^{-14}$
	$N = 240$	$\Delta n = 0$	-4.22759×10^{-4}	$\Delta \mathcal{U}/\mathcal{U} = 3.952 \times 10^{-14}$
Drive	$\Delta t = 0.020$	5.333×10^{-2}	4.538×10^{-6}	$\Delta Q/Q = 1.478 \times 10^{-13}$
	$\Delta t = 0.010$	5.351×10^{-2}	1.135×10^{-6}	$\Delta Q/Q = 2.376 \times 10^{-13}$
	$\Delta t = 0.005$	5.355×10^{-2}	2.837×10^{-7}	$\Delta Q/Q = 8.704 \times 10^{-13}$

C Artifact and number traceability

The implementation is in `code/lattice.qed.py`; production entry points are `run_scope.py`, `run_boundary.py`, and `run_reproductions.py`. The immutable prediction protocol is `code/protocol.json`. Human-readable summaries end in `_summary.json`; the corresponding NPZ files contain modal observables and full conservation histories. The paper-reproduction NPZ files additionally contain every stored time level of ϕ_v and all nonzero A_e components for seed 1729. Temporal A_0 and laser A_z are identically zero and are stated as such in their configurations.

`code/derive_predictions.py` independently recomputes analytic frequencies, effective-action coefficients, and all macros in `logs/numbers.tex`; `logs/paper_numbers.json` maps each macro to its source summary. `code/analyze_scope_overlays.py` derives the two parameter-free comparison curves from the original modal histories and writes `logs/scope_overlay_trajectories.npz` together with checksums and full-record errors in `logs/scope_overlay_metrics.json`. Figures are regenerated by `code/make_figures.py`. Development approaches that were rejected before production are recorded in `attempts.log.md`.

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